

TOI-1516b

R-2136

Benchmark run · workflow W8-benchmark-deep · record deep-bench_TOI1516_250826 · created 2026-08-11T23:54:04+00:00

PRACTICE / VALIDATION RUN — not submitted to any science database

Results

Mid-Transit Time (Tmid)	2460913.921 +/- 0.044 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.162 +/- 0.047
Transit depth (Rp/Rs)^2	2.61 +/- 1.52 [%]
Orbital Inclination (inc)	89.68 +/- 1.44
Airmass coefficient 1 (a1)	0.192 +/- 0.027
Airmass coefficient 2 (a2)	1.25 +/- 0.74
Scatter in the residuals of the lightcurve fit is	14.9 %
Best Comparison Star	#2 - [539, 323]
Optimal Aperture	3.23
Optimal Annulus	7.38
Transit Duration (day)	0.085 +/- 0.032

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.162 ± 0.047 vs 0.1224 (deviation 0.84σ)
Duration fit vs published	0.085 d vs 0.1217 d (deviation 1.15σ)
Mid-time O–C	87.6 min
Depth SNR	3.4
Residual scatter	14.9 %

Light curve

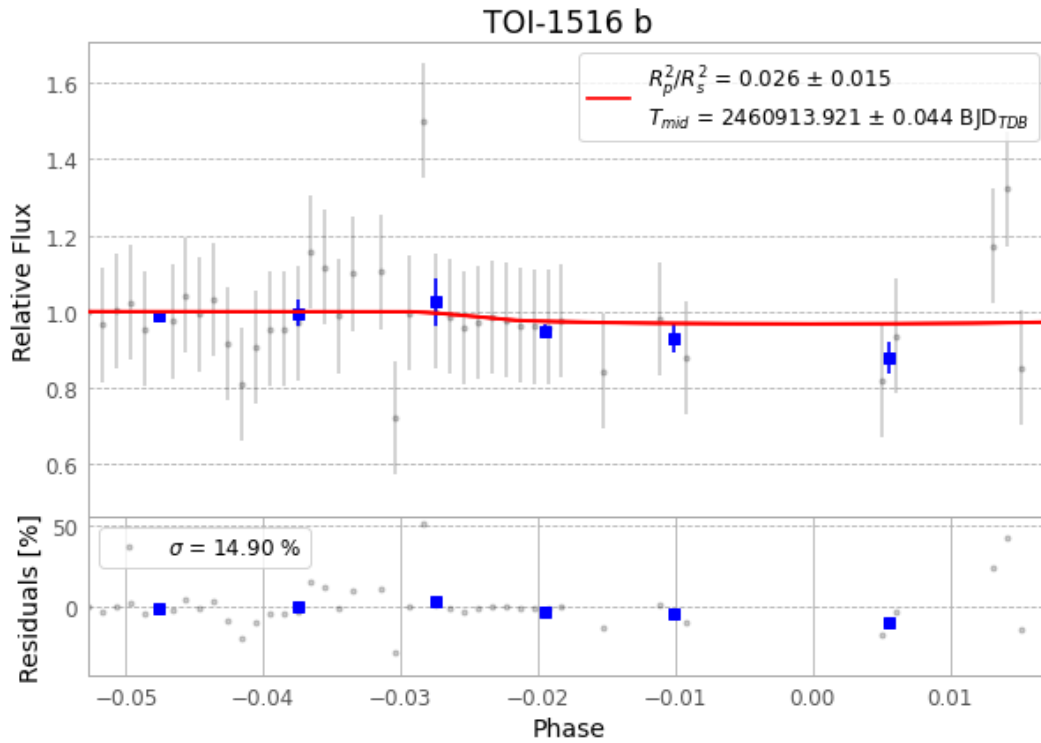


Figure 1 — FinalLightCurve_TOI-1516 b_2025-08-26.png (SHA-256 13124438b08f4cb1...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [398, 286], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 86f2c8943ff34b18...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit fd74a91

Data provenance

72 FITS frames (47 MB), TOI1516250826072715.FITS → TOI1516250826110015.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-TOI1516-250826.log (b4ec810e31c6...), FinalLightCurve_TOI-1516 b_2025-08-26.png (13124438b08f...), FinalLightCurve_TOI-1516 b_2025-08-26.csv (01c600c1bf00...)

Compute resources

Wall time 197.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-11 23:54Z.